

DSA test 2

Question 1

```
class First{
    public static void main(String[] args) {
        System.out.println("Hello, World!");

        int n=4;
        System.out.println("Factorial of "+n+" is: "+ factorial(n));

    }
    static int factorial(int n){
        if(n==0 || n==1){
            return 1;
        }
        return n*factorial(n-1);
    }
}
```

Question 2:

```
class Second{
    public static void main(String[] args) {
        System.out.println("This is the Second Java file.");
        int a=0;
        int b=1;
        for(int row=1;row<=3;row++){
            for(int col=1;col<=row;col++){

                int c=a+b;
                System.out.print(c+" ");
                a=b;
                b=c;
            }
            System.out.println();
        }
    }
}
```

Question 3:

```
class Third {
    public static void main(String[] args) {
        System.out.println("This is the Third class.");
        for(int i=3;i>=1;i--) {
            for(int j=1;j<=i;j++) {
                System.out.print("*");
            }
            System.out.println();
        }
        System.out.println();

        for(int i=3;i>=1;i--) {

            for(int s=2;s<=i;s++) {
                System.out.print(" ");
            }
        }
    }
}
```

```

        for(int j=3;j>=i;j--) {
            System.out.print(j);
        }
        for(int k=i+1;k<=3;k++) {
            System.out.print(k);
        }
        System.out.println();
    }
    for(int i=2;i<=3;i++) {

        for(int s=1;s<i;s++) {
            System.out.print(" ");
        }

        for(int j=3;j>=i;j--) {
            System.out.print(j);
        }
        for(int k=i+1;k<=3;k++) {
            System.out.print(k);
        }
        System.out.println();
    }
}

```

Question 4:

```

import java.util.Scanner;

class Fourth {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("This is the Fourth class.");
        System.out.println("Enter a number:");
        int n=sc.nextInt();
        binaryreverse(n);
    }
    static void binaryreverse(int n){
        while(n>0) {
            int bin=n%2;
            System.out.print(bin);
            n=n/2;
        }
    }
}

```